

RENANCE

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# FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

PEER 314 — Past Questions & Materials  
**by Renance**

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Course: PEER 314  
Institution: Federal University of Petroleum Resources, Effurun  
Papers in this pack: 1  
Levels: 300

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# Paper 1 of 1 — PEER 314 Petroleum Geology 2025

Level 300 • First Semester • PastQuestion

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COE 11177

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN  
COLLEGE OF ENGINEERING AND TECHNOLOGY  
DEPARTMENT OF PETROLEUM ENGINEERING  
PROGRAMME: B.ENG IN PETROLEUM ENGINEERING  
SESSION: 2024/2025. SEMESTER: FIRST SEMESTER EXAMINATION

|               |                   |               |                |
|---------------|-------------------|---------------|----------------|
| COURSE TITLE: | PETROLEUM GEOLOGY | COURSE UNIT:  | 3              |
| COURSE CODE:  | PEE 314           | DATE OF EXAM: | 17th MAY, 2025 |

## SECTION A (Answer any 2 questions)

1. a. What is a reservoir rock? (5 Marks)  
b. Write short notes on the physical properties of a reservoir. (10 Marks)
2. a. Briefly discuss the concept of petroleum migration. (5 Marks)  
b. Define porosity and differentiate between effective and total porosity using a diagram. (10 Marks)
3. Write short notes on the following:
  - i. Fracture porosity. (3 Marks) *Crack in the pores / breaking rocks*
  - ii. Sorting. (3 Marks) *Proportion and distribution of grains in the grain matrix*
  - iii. Compaction. (2 Marks) *Fine grained sediments under pressure*
  - iv. Cementation. (2 Marks) *This can be defined as lithification of*
  - v. Water saturation. (5 Marks) *Initial*

## SECTION B (Answer all questions in this section)

- 4(a). (i) What are the processes followed by oil and gas companies in the production and exploration of petroleum/crude oil? (5 Marks)  
(ii) What is the origin of petroleum, and what does the term "petroleum" mean? (2 Marks)  
(iii) List five logging tools commonly used in the oil and gas industry and the parameters each measure. (10 Marks)
- 4(b) (i) Provide some examples (3) of reservoir rock. (3 Marks)  
(ii) Briefly explain how crude oil is found and extracted from a reservoir rock. (5 Marks)

## SECTION C (Answer only one question)

- 5(a). Define reservoir trap and state two functions of a reservoir trap. (5 Marks)
- (b). State two examples of each of the following traps, providing well-labeled diagrams:
  - i. Structural trap
  - ii. Stratigraphic trap (10 Marks)
- 6(a). Calculate the stock tank oil initially in place and the gas initially in place given the following parameters: Area = 2,000 acres; Thickness = 40 ft; Porosity = 15%; Initial water saturation = 20%; Initial oil formation volume factor = 1.23 bbl/stb; Initial gas formation volume factor = 0.0075 cf/scf. (8 Marks)
- (b). A reservoir has a pore volume of 750 cu ft, with an oil saturation of 15%. What is the volume of oil occupying the pore space? (7 Mark)